



OUR PRODUCTS DEVELOP TOMORROW'S TECHNOLOGIES™

AUTOEXPLOR™

IP CLIENT USER MANUAL



For All
EXPLORAVAC™
Systems

AEX-IPC-11012024 - V 1.4.0.0

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1. Introduction

This document is intended to help advanced users to start building client software to interface with *AutoExplor*. Unlike with the [AutoExplor API](#), clients cannot access the *ExploraVac* instrument directly. *AutoExplor* is required to act as the server. The structure and commands of the *AutoExplor IP Client* and *AutoExplor API* are largely the same.

2. Getting Started

Before getting started, make sure that:

1. *AutoExplor* is running and connected to the *ExploraVac*. The IP Client feature can only be used with a Premium key.
2. *AutoExplor* is set to allow for incoming connections. Upon installing *AutoExplor*, this setting is disabled. To enable this feature go to: “Settings → System Information”, and click the “Allow Incoming Connections” button. Upon checking this for the first time, your firewall will typically pop up and request *AutoExplor* be given an exception to communicate on your network. This exception must be allowed in order for *AutoExplor* to accept incoming connections.
3. Your network setup allows incoming connections between the computer you are connecting from and the computer you are connecting to. Typically, this means that your computer should be on the same domain as the one you are connecting to. Connections from a computer to itself are always allowed.

Once these steps are complete you are ready to build a client for *AutoExplor*. If the provided sample code is not working at this stage, contact your IT department for expert help on diagnosing the problem. It is likely a firewall issue. If your IT department requires additional assistance, please contact Ideal Vacuum.

3. Connecting to AutoExplor

All application layer communication is done with UTF-8 encoding. Please see the included working client code samples.

3.1 Required - AutoExplor Connection Port

To connect to *AutoExplor*, a TCP connection must be made. By default, *AutoExplor* listens on port 23513*. Finding the IP address of your host computer can either be hard-coded or done through the discovery port (described below).

3.2 Optional - AutoExplor Discovery Port

To find *AutoExplor* instances on your network, send a broadcast datagram with the string “AutoExplor Discovery” via UDP to port 23514* (default). This will cause all instances on your network to respond with their “SerialNumber”, “PCName”, and “AutoExplor Version”, delimited by spaces. From the IP address of the response, you should then be able to connect to *AutoExplor* as outlined above.

**In the rare case that either of these two ports are used by another program on your computer, these port settings can be changed by going to the AutoExplor install folder -> Resources -> NetworkSettings.xml. Change the ConnectionPort and/or DiscoveryPort attributes to an unused port on your machine.*

4. Loading the ExploraVac Configuration

Once you have connected to *AutoExplor*, you can now look at its configuration. The configuration tells you what subsystems are on the connected *ExploraVac*. The four categories below correspond to the different *ExploraVac* subsystem types and are grouped by their behavior. Each bit of a category corresponds to one distinct module. Below, in Section 4.1, Tables 1 – 4 denote the component that correlates to each bit.

Device (DEVICEIDS): Read-only properties that detail the current state of each device on the *ExploraVac*.

Readout (READOUTIDS): Simple read-only floating-point value that also contains a source unit.

Button (BUTTONIDS): Simple on/off control. These correspond to the physical buttons on the control panel or the virtual buttons in *AutoExplor*.

Setpoint Controller (SETPOINTIDS): A complex control that attempts to keep the linked value at the same value as its target setpoint. Also includes ramping and soaking functionality.

Code	Description
VENT VALVE (1)	Vent valve properties.
GATE VALVE (2)	Gate valve properties.
ROUGH VALVE (4)	Roughing line valve properties.
T CUBE (8)	TCube solid state cooling system properties.
TURBO WATER CHILLER (16)	Liquid-cooled turbo chiller properties.
ROUGH PUMP (32)	Roughing pump properties.
CRYO PLATEN PUMP (64)	Coolant pump properties (ethanol cooled systems).
COMPRESSOR (128)	Air compressor properties.
TURBO (256)	Turbo pump properties.
PURGE VALVE (512)	Purge valve properties.
CHAMBER HEATER (1024)	Chamber heating properties
PLATEN HEATER (2048)	Platen heating properties.
VAPOR TRAP (4096)	Vapor trap bath properties (ethanol cooled systems).
CHAMBER SYSTEM (8192)	Overall system properties.
LN2 SYSTEM (16384)	Liquid nitrogen powered cooling system properties.
H/C RECIRCULATOR (32768)	Recirculating heating & cooling system.

Table 1: DEVICEIDS

Code		Description
CHILLER BATH	(8)	Temperature of bath (ethanol cooled systems).
PLATEN TC	(16)	Temperature of the platen thermocouple (attached under platen)
SAMPLE 1	(32)	Optional thermocouple readout (see ExploraVac manual for hookup).
PRESSURE	(64)	Pressure readout on non-pressure-controlled systems.
CUSTOM RO 1	(128)	Custom ExploraVac readout (see ExploraVac manual, Additional Readouts)
CUSTOM RO 2	(256)	Custom ExploraVac readout (see ExploraVac manual, Additional Readouts)
CUSTOM RO 3	(512)	Custom ExploraVac readout (see ExploraVac manual, Additional Readouts)
SAMPLE 2	(1024)	Optional thermocouple readout (see ExploraVac manual for hookup).
SAMPLE 3	(2048)	Optional thermocouple readout. (See ExploraVac manual for hookup)
SAMPLE 4	(4096)	Optional thermocouple readout. (See ExploraVac manual for hookup)
SAMPLE 5	(8192)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 6	(16384)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 7	(32768)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 8	(65536)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 9	(131072)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 10	(262144)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 11	(524288)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 12	(1048576)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 13	(2097152)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 14	(4194304)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 15	(8388608)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 16	(16777216)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 17	(33554432)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 18	(67108864)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 19	(134217728)	Optional thermocouple readout (optional thermocouple package required).
SAMPLE 20	(268435456)	Optional thermocouple readout (optional thermocouple package required).

Table 2: READOUTIDS

Code	Description
POWER (1)	On/Off switch for the system.
HIGH VAC (2)	On/Off switch for high vacuum mode.
VENT (4)	On/Off switch for vent mode.
PURGE (128)	On/Off switch for purge mode.
LIGHT (256)	On/Off switch for chamber lighting.
CHILLER (512)	On/Off switch for prechilling the ethanol bath.
ROUGH (1024)	On/Off switch for rough vacuum mode.
AUX 1 (2048)	On/Off relay for auxiliary port 1 (see <i>ExploraVac</i> manual for hookup).
AUX 2 (4096)	On/Off switch for auxiliary port 2 (see <i>ExploraVac</i> manual for hookup).
CUSTOM_BUTTON1 (8192)	Custom Button. See <i>ExploraVac</i> manual appendix.
CUSTOM_BUTTON2 (16384)	Custom Button. See <i>ExploraVac</i> manual appendix.
CUSTOM_BUTTON3 (32768)	Custom Button. See <i>ExploraVac</i> manual appendix.
CUSTOM_BUTTON4 (65536)	Custom Button. See <i>ExploraVac</i> manual appendix.
CUSTOM_BUTTON5 (131072)	Custom Button. See <i>ExploraVac</i> manual appendix.

Table 3: *BUTTONIDS*

Code	Description
CHAMBER HEATING CONTROL (8)	Chamber wall heating setpoint control.
PLATEN HEATING CONTROL (16)	Platen heating/cooling setpoint control.
PRESSURE CONTROL (64)	Roughing range setpoint pressure control.
CUSTOM_SP1 (262144)	Custom setpoint. See <i>ExploraVac</i> manual appendix.
CUSTOM_SP2 (524288)	Custom setpoint. See <i>ExploraVac</i> manual appendix.

Table 4: *SETPOINTIDS*

5. Querying Data and Setting Controls on ExploraVac

5.1 General Information

All commands to and from the server should be in UTF-8 encoding and be followed up with a line feed (CR+LF is also acceptable). Responses are delimited by carriage return + line feed (CR+LF). When reading a response there are two parts delimited by a semicolon. The first value is the error code found in Table 5, below. The second value is the response body, typically delimited by spaces for multiple value responses.

Each command has an associated permissions level. There are two permission levels: “Read Only” and “Full Control”. The default permissions level when a client connects is “Read Only”. A client with “Read Only” permissions is only allowed to view data from the *ExploraVac* and cannot manipulate the device in any way. Any number of clients are allowed to have “Read Only” permissions. Only the host or a single connected client is allowed to have “Full Control”. A client/host with “Full Control” permissions is allowed to manipulate the *ExploraVac* freely, start recipes, etc. If a client with “Full Control” disconnects, the host takes back “Full Control” privileges. Clients cannot request “Full Control” while another client has control but can while the host has control. The host machine can always take back control by going to “Settings → System Information” and clicking the “Take Control” button. Additionally, the host machine can forcefully disconnect a client if desired from the “Settings → System Information” page. In this way, *AutoExplor* directly connected to the *ExploraVac* will never be denied from controlling the *ExploraVac* to which it is connected. The physical button panel and the emergency stop are not affected by *AutoExplor*.

5.2 Responses

All commands have a unified pattern. Every response from the host has a status delimited by a semicolon then a response body. In addition, each command ends with CR+LF. Status Codes are found below in Table 5.

Response Pattern: {STATUSCODE};{RESPONSEBODY}\r\n

STATUSCODE: Status of the command executed. (See below.)

RESPONSEBODY: The data from the command. (See command response for details per response)

Note: A Success (0) code does not guarantee the desired effect was achieved. Rather, it indicates that the command was attempted. This may cause confusion and a source of bugs in client code. Many factors can cause a module to switch state, from errors to system state. It is good practice to continually check the status of a particular module to verify it is behaving as expected.

Code	Description
Success (0)	Command executed without issue.
Not Run (1)	Command not executed for some reason.
Not Allowed (2)	Command not allowed to be run, due to configuration or state of system.
Incorrectly Formed (3)	Command could not be executed due to improper syntax.
Permissions Error (4)	Command needs elevated permissions (Full Control) before executing.
Disconnected (5)	<i>ExploraVac</i> disconnected from the host.
Server Error (6)	Unexpected error occurred while executing the command.
Not Recognized (7)	Command not recognized by the server.

Table 5: STATUSCODES

5.3 Keep Alive Packets

To handle unexpected disconnects, the server requires every client to make a request every 10 minutes. If a client remains silent for 10 minutes after its last request (including incorrectly formed requests), the client will be disconnected by the host. Keep this in mind if you have a lightweight process.

5.4 Configuration Commands

5.4.1 GetDeviceConfiguration

Usage: GetDeviceConfiguration Response Body: {CONFIG}
Summary: Gets the device configuration of the system.
Permissions: Read Only.
CONFIG: Returns the device configuration of the system (see Table 1: DEVICEIDS).

5.4.2 GetReadoutConfiguration

Usage: GetReadoutConfiguration Response Body: {CONFIG}
Summary: Gets the readout configuration of the system.
Permissions: Read Only.
CONFIG: Returns the readout configuration of the system (see Table 2: READOUTIDS).

5.4.3 GetButtonConfiguration

Usage: GetButtonConfiguration Response Body: {CONFIG}
Summary: Gets the button configuration of the system.
Permissions: Read Only.
CONFIG: Returns the button configuration of the system (see Table 3: BUTTONIDS).

5.4.4 GetSetpointConfiguration

Usage: GetSetpointConfiguration Response Body: {CONFIG}
Summary: Gets the setpoint configuration of the system.
Permissions: Read Only
CONFIG: Returns the setpoint configuration of the system (see Table 4: SETPOINTIDS).

5.5 Device Commands

5.5.1 GetDeviceName

Usage: GetDeviceName {DEVICEID} Response Body: {NAME}
Summary: Gets a friendly name for the device.
Permissions: Read Only.
DEVICEID: Device Identifier.
NAME: Name of the device.

5.5.2 GetDeviceState

Usage: GetDeviceState {DEVICEID} Response Body: {ON/OFF}
Summary: Gets the state for the device.
Permissions: Read Only.
DEVICEID: Device Identifier.
ON/OFF: 1 = On, 0 = Off

5.5.3 GetDeviceProperties

System devices vary between *ExploraVac* configurations, and each system device has different properties. For example, the device properties of one model of turbo pump are different than the device properties of another model. GetDeviceProperties gets all the different properties for a device and outputs them in comma-delimited keys. These keys are space and case sensitive. Use these keys in GetDeviceProperty to get the value for that particular property.

Usage: GetDeviceProperties {DEVICEID} Response Body: {DATA}

Summary: Gets the comma-delimited names for each property on the device.

Permissions: Read Only

DEVICEID: Device Identifier.

DATA: Comma-delimited string that contains the properties of the device (there may be spaces).

Example mapping from CHAMBER SYSTEM:

“System Current , Cabinet Temp , Ion Gauge” →

“System Current “ , “Cabinet Temp “ , “Ion Gauge”

5.5.4 GetDeviceProperty

Usage: GetDeviceProperty {DEVICEID} “{PROPERTYNAME}” Response Body: {DATA}

Summary: Gets the property value and tag (if applicable) for the device.

Permissions: Read Only.

DEVICEID: Device Identifier.

PROPERTYNAME: Property from which to retrieve the value. Use double quotes around the key.

DATA: Value of the property with unit tag appended (i.e., “System Current “ → 10.2Amps)

5.5.5 GetDevicePropertyNoUnits

Usage: GetDevicePropertyNoUnits {DEVICEID} “{PROPERTYNAME}” Response Body: {DATA}

Summary: Gets the property value for the device.

Permissions: Read Only.

DEVICEID: Device Identifier.

PROPERTYNAME: Property from which to retrieve the value. Use double quotes around the key.

DATA: Value of the property, no units (i.e., “System Current “ → 10.2)

5.6 Button Commands

5.6.1 GetButtonName

Usage: GetButtonName {BUTTONID} Response Body: {NAME}

Summary: Gets a friendly name for the button.

Permissions: Read Only.

BUTTONID: Button Identifier.

NAME: Name of the button.

5.6.2 GetButtonState

Buttons and setpoints have a “colorStatus” field. Color states represent actual operation of the control. A control can be “on” but in a startup sequence (BLINK GREEN). In this “BLINK GREEN” example, the color represents a startup sequence, whereas the “On/Off” field represents the desire for the control to be on. See the definitions section at the end of this section and the section on [warnings and other subsystem conditions, P. 17](#), for a table of color state indicators for buttons and setpoints.

Usage: GetButtonState {BUTTONID} Response Body: {ON/OFF} {COLORSTATE}

Summary: Gets the state of the button.

Permissions: Read Only.

BUTTONID: Button Identifier.

ON/OFF: 1 = On, 0 = Off

COLORSTATE: Color state of the setpoint control. Typically, 0 (None) = off, 1 (Solid Green) = on.

Others are detailed at the end of this section and in [Sec. 3.7, p. 17](#).

5.6.3 SetButtonState

Usage: SetButtonState {BUTTONID} {ON/OFF} Response Body:

Summary: Tries to set the button to desired state.

Permissions: Full Control.

BUTTONID: Button Identifier.

ON/OFF: 1 = On, 0 = Off

5.7 Readout Commands

5.7.1 GetReadoutName

Usage: GetReadoutName {READOUTID} Response Body: {NAME}

Summary: Gets a friendly name for the readout.

Permissions: Read Only.

READOUTID: Readout Identifier.

NAME: Name of the readout.

5.7.2 GetReadoutNativeUnit

Usage: GetReadoutNativeUnit {READOUTID} Response Body: {UNIT}

Summary: Gets the native unit for the readout (Torr / °C typically).

Permissions: Read Only.

READOUTID: Readout Identifier.

UNIT: Native unit in which this readout operates.

5.7.3 GetReadoutProcessValue

Usage: GetReadoutProcessValue {READOUTID} Response Body: {VALUE}

Summary: Gets the value of the readout in native units.

Permissions: Read Only.

READOUTID: Readout Identifier.

VALUE: Value of the readout. NaN indicates invalid data (sensor may be off or starting up).

5.8 Setpoint Commands

5.8.1 GetSetpointName

Usage: GetSetpointName {SETPOINTID} Response Body: {NAME}

Summary: Gets a friendly name for the setpoint.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

NAME: Name of the setpoint.

5.8.2 GetSetpointNativeUnit

Usage: GetSetpointNativeUnit {SETPOINTID} Response Body: {UNIT}

Summary: Gets the native unit for the setpoint (Torr / °C typically).

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

UNIT: Native unit in which this setpoint operates. This does not change when swapping to alt(ft.).

5.8.3 GetSetpointIsInAltitudeModeState

Because altitude works in a non-linear fashion, this function exists to assist users in setting setpoints and ramp rates. This mode persists on *ExploraVac* and will remain even if you restart *AutoExplorAPI*. This mode only exists for pressure.

Usage: GetSetpointIsInAltitudeModeState {SETPOINTID} Response Body: {YES/NO}

Summary: Gets if this is in altitude mode.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

YES/NO: 1 = alt(ft.) mode, 0 = standard (Torr) mode

5.8.4 GetSetpointCanDoAltitude

Usage: GetSetpointCanDoAltitude {SETPOINTID} Response Body: {YES/NO}

Summary: Gets if this setpoint can be put into altitude mode.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

YES/NO: 1 = Can do Altitude Mode, 0 = Cannot do Altitude Mode.

5.8.5 GetSetpointState

Buttons and setpoints have a “colorStatus” field. Color states represent actual operation of the control. A control can be “on” but in a startup sequence (BLINK GREEN). In this “BLINK GREEN” example, the color represents a startup sequence, whereas the “On/Off” field represents the desire for the control to be on. See the definitions section at the end of this section and the [warnings and other subsystem conditions, P. 17](#) for a chart on color state indicators for buttons and setpoints.

Usage: GetSetpointState {SETPOINTID} Response Body: {ON/OFF} {COLORSTATE}

Summary: Gets the current status of the setpoint controller

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

ON/OFF: 1 = On, 0 = Off

COLORSTATE: Color state of the setpoint control. Typically, 0 (None) = off, 1 (Solid Green) = on.

Others are detailed at the end of this section and in [Sec. 3.7, p. 17](#).

5.8.6 GetSetpointProcessValue

Usage: GetSetpointProcessValue {SETPOINTID} Response Body: {VALUE}

Summary: Gets the native value that this setpoint controller is monitoring.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

VALUE: The sensor value in native units. Always native unit regardless of altitude mode.

5.8.7 GetSetpointTargetValue

Usage: GetSetpointTargetValue {SETPOINTID} Response Body: {VALUE}

Summary: Gets the setpoint target value.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

VALUE: The value of the setpoint in native units. Always native unit regardless of altitude mode.

5.8.8 GetSetpointTTV

This is the value that the *ExploraVac* is attempting to reach at any given moment (temperature or pressure). When a ramp rate has not been specified, the SetpointTTV value equals the user entered setpoint value.

When a ramp rate (temperature or pressure) has been specified, the SetpointTTV will be different than the user entered target setpoint value. The SetpointTTV will approach the target setpoint until it is reached.

Example: You are ramping from 20°C to 30°C at a ramp rate of 2°C per minute. After the first minute, the TTV will be 22°C. At two minutes, the TTV will be 24°C, and so on until the entered setpoint value is reached. Meanwhile, the user entered target setpoint value remains constant at 30°C.

Usage: GetSetpointTTV {SETPOINTID} Response Body: {VALUE}

Summary: Gets the Transient Target Value (TTV) of the setpoint.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

VALUE: The value of the TTV in native units. Always in native units regardless of altitude mode.

5.8.9 GetSetpointRampValue

Usage: GetSetpointRampValue {SETPOINTID} Response Body: {VALUE}

Summary: Gets the native units per minute ramp rate of the control. In altitude mode, this is in ft./min.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

VALUE: The ramp rate in native units or ft./min. depending on mode. 0 = Max Ramp Rate

5.8.10 GetSetpointSoakValue

Usage: GetSetpointSoakValue {SETPOINTID} Response Body: {VALUE}

Summary: Gets the soak duration in seconds of the control.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

VALUE: The soak duration in seconds. 0 = No soak

5.8.11 GetSetpointModeOptions

Modes exist on some setpoint controllers based on the system's configuration. These modes are comma delimited. See the *ExploraVac* manual for more details on modes and their operation.

Example: A system with purge lines and a pressure setpoint controller will have vent, purge, and active purging (actprg) options available from which to select. Alternatively, a system without purge lines will not have any mode options available, as it can only vent.

Usage: GetSetpointModeOptions {SETPOINTID} Response Body: {DATA}

Summary: Gets the comma delimited operating setpoint modes, or empty string if no modes present.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

DATA: Comma-delimited string containing the setpoint modes, or empty string if no modes present.

Example of possible PRESSURE CONTROL: "vent,purge,actprg"-> "vent", "purge", "actprg"

5.8.12 GetSetpointMode

Usage: GetSetpointMode {SETPOINTID} Response Body: {MODE}

Summary: Gets the current mode of the setpoint controller, or empty string if no modes present.

Permissions: Read Only.

SETPOINTID: Setpoint Identifier.

MODE: String that indicates the current mode of the setpoint.

5.8.13 SetSetpointTargetValue

Usage: SetSetpointTargetValue {SETPOINTID} {VALUE} {FROMVALUE} Response Body:

Summary: Sets the setpoint target of the controller.

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

VALUE: This is the value that the setpoint controller will ramp to.

FROMVALUE: (Optional) If FROMVALUE is not used, this will ramp from the current value. If it is used, it will ramp from the value provided.

5.8.14 SetSetpointRamp

Usage: SetSetpointRamp {SETPOINTID} {VALUE} Response Body:

Summary: Sets the setpoint ramp rate.

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

VALUE: The value to set the ramp rate in native units. If altitude mode is active, use alt(ft.) instead.

5.8.15 SetSetpointSoak

Usage: SetSetpointSoak {SETPOINTID} {VALUE} Response Body:

Summary: Sets the setpoint soak duration.

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

VALUE: The value to set the soak duration in whole seconds. 0 = no soak.

5.8.16 SetSetpointIsInAltitudeModeState

Usage: SetSetpointIsInAltitudeModeState {SETPOINTID} {ON/OFF} Response Body:

Summary: Sets the mode to value given. See GetSetpointIsInAltitude above section for more details on how this special mode functions.

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

ON/OFF: 1 = alt(ft.) mode, 0 = standard (Torr) mode

5.8.17 SetSetpointState

Usage: SetSetpointState {SETPOINTID} {ON/OFF} Response Body:

Summary: Queues up setting the button to the desired state on the next call to Update().

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

ON/OFF: 1 = On, 0 = Off

5.8.18 SetSetpointMode

Usage: SetSetpointMode {SETPOINTID} {MODE} Response Body:

Summary: Queues up setting the button to the desired state on the next call to Update().

Permissions: Full Control.

SETPOINTID: Setpoint Identifier.

MODE: String that indicates the current mode of the setpoint.

5.8 Global Commands

5.9.1 RequestControl

Usage: RequestControl Response Body:

Summary: Requests full control from the host. If available this returns a success.

Permissions: Read Only.

5.9.2 RelinquishControl

Usage: RelinquishControl Response Body:

Summary: Downgrades client to "Read Only" privileges and give back control to the host.

Permissions: Full Control.

5.9.3 StartRecipe

Usage: StartRecipe {PATH} Response Body:

Summary: Starts a recipe on the host system.

Permissions: Full Control.

PATH: Absolute path of the recipe file FROM THE HOST. Alternatively, attempts to find via a local path from the default recipe directory on the host.

5.9.4 StopRecipe

Usage: StopRecipe Response Body:

Summary: Stops a recipe on the host system.

Permissions: Full Control.

5.9.5 IsRecipeRunning

Usage: IsRecipeRunning Response Body: {YES/NO}
Summary: Checks to see if a recipe is running on the system.
Permissions: Read Only.
YES/NO: 1 = Yes, 0 = No

5.9.6 StreamRecipe

Usage: StreamRecipe {BEEPBUZZER} {STARTSTEP} {NUMBEROFRUNS} {STREAMLENGTH}
Response Body:
Summary: Streams and starts an exported .recipe file of size StreamLength bytes along the network.
Permissions: Full Control.
BEEPBUZZER: Will beep buzzer when recipe completes. 1 = Yes, 0 = No
STARTSTEP: Recipe step to start at.
NUMBEROFRUNS: Number of runs the recipe will do.
STREAMLENGTH: The number of subsequent bytes (after newline) that will be read as a recipe.

5.9.7 GetRecipeStatusUpdate

Usage: GetRecipeStatusUpdate
Response Body: {STATE} [{RECIPESTEP} {TIMESEXECUTED} {REECIPECOUNT}]
STATE: State of the running recipe. 0 = No Recipe / Completed Successfully, 1 = Running, 2 = Stopped due to error, 3 = Stopped due to user input.
RECIPESTEP: (Only if running) Step that the recipe is currently executing.
TIMESEXECUTED: (Only if running) Number of times a step has currently run.
REECIPECOUNT: (Only if running) The number of times a recipe has been run. 0 = First time.

5.9.8 GetRunningRecipeStream

Usage: GetRunningRecipeStream
Response Body: {BEEPBUZZER} {CURRENTSTEP} {TIMESEXECUTED} {NUMBEROFBYTES}
Summary: Streams the recipe file on the host with NumberOfBytes following this response
Permissions: Read Only
BEEPBUZZER: Will beep buzzer when recipe completes. 1 = Yes, 0 = No
RECIPESTEP: Step that the recipe is currently executing.
TIMESEXECUTED: Number of times a step has currently run.
NUMBEROFBYTES: The number of bytes in the subsequent stream.

5.9.9 HasControl

Usage: HasControl Response Body: {PERMISSIONS}
Summary: Tells the client if they have Full Control / Read Only permissions.
Permissions: Read Only
PERMISSIONS: 1 = Full Control, 0 = Read Only

5.9.10 SkipRecipeStep

Usage: SkipRecipeStep {SKIPINDEX} Response Body:
Summary: Forces the recipe to move on along recipe "AND block" index.
Permissions: Full Control
SKIPINDEX: AND index though which the recipe will move on though.

5.9.11 GetExploraVacMetadata

Usage: GetExploraVacMetadata

Response Body: {SNUM} {MODEL} {DOM} {EVVERSION} {AEVERSION}

Summary: Gets basic supporting information for ExploraVac and the AutoExplor host.

Permissions: Read Only

SNUM: Serial Number of the ExploraVac connected to the AutoExplor host.

MODEL: Model information of the ExploraVac connected to the AutoExplor host.

DOM: Date of Manufacture of the ExplorVac connected to the AutoExplor host.

EVVERSION: Version number of the firmware of the ExploraVac connected to the AutoExplor host.

AEVERSION: Version number of the AutoExplor host.

5.9.12 SilenceAlarms

Usage: SilenceAlarms

Response Body:

Summary: Silences alarm until errors change or powered off

Permissions: Full Control

5.10 Definitions

Code	Description
NONE (0)	Device control is off. Physical button panel is SOLID RED.
SOLID GREEN (1)	Devices control is on and running properly.
SOLID RED (2)	Reserved.
SOLID YELLOW (3)	Reserved.
SOLID BLUE (4)	see Sec. 3.7, p. 17
BLINK RED (5)	see Sec. 3.7, p. 17
BLINK GREEN (6)	see Sec. 3.7, p. 17
GREEN YELLOW (7)	see Sec. 3.7, p. 17
GREEN RED (8)	see Sec. 3.7, p. 17
RED YELLOW (9)	see Sec. 3.7, p. 17
BLINK BLUE (10)	Reserved.

Table 6: COLORSTATES

6. Error, Warning, and Service Notifications

Error, warning, and service handling follow the same scheme as buttons, readouts, devices, and setpoints. Each is an enumerated value with a bit representing if that error is active or not. Services have one special facet in that you likely need to mark it as serviced when you have completed your maintenance. For maintenance items such as tip seal replacement for a pump where there is no detection on change of the system, an estimated lifetime needs to be reset on the pump. Notification values can be queried using the functions listed in Tables 7, 8, and 9, below. Some notifications can occur while the system is off, but most occur only when the system is on.

6.1 Errors

6.1.1 GetExploraVacSystemErrors

Usage: GetExploraVacSystemErrors Response Body: {CODES}

Summary: Gets the current errors on the system. Use ERRORDEFINITIONS to decode.

Code	Description
CHAMBER OVER TEMP (1)	Chamber temperature more than failure threshold.
TC CHAMBER BURNOUT (2)	Chamber thermocouple unplugged or failed.
PLATEN OVER TEMP (4)	Platen temperature more than failure threshold.
TC PLATEN BURNOUT (8)	Platen thermocouple unplugged or failed.
CHAMBER OVER PRESSURE (16)	Gauges detecting an overpressure condition inside the chamber.
ROUGH PUMP ERROR (32)	Roughing pump error detected. See device properties.
NO TANK PRESSURE (64)	Compressor low or zero air pressure.
TC PUMP FLOW FAULT (128)	TCube chiller failed due to a pump flow fault or blockage.
ETH PUMP STUCK (256)	Ethanol pump relay sticking or failing.
OXY LVL CRITICAL (512)	LN2 system detected unsafe atmospheric oxygen levels.
TURBO COM FAIL (1024)	Communication with turbo pump disrupted.
ROUGH COM FAILED (2048)	Communication with roughing pump disrupted.
LN2 HEAT FAILURE (4096)	LN2 exhaust temperature more than 200°C. LN2 system shutdown.
LN2 EX TC BURNOUT (8192)	LN2 exhaust thermocouple unplugged or failed.
OUT OF LN2 (16384)	LN2 fully depleted. LN2 system shutdown.
RECIRCULATOR ERROR (32768)	View heating & cooling recirculator device properties for error code.

Table 7: Error Codes

CODES: The current error state of the system.

6.2 Warnings

6.2.1 GetExploraVacSystemWarnings

Usage: GetExploraVacSystemWarnings Response Body: {CODES}

Summary: Gets the current warnings on the system. Use WARNINGDEFINITIONS to decode.

Code	Description
CHAMBER OVER TEMP (1)	Chamber temperature more than warning threshold.
LN2 LOW (2)	LN2 system tanks nearly empty.
PLATEN OVER TEMP (4)	Platen temperature more than warning threshold.
GATE VALVE N OPEN (8)	Cannot detect or verify gate valve is open.
GATE VALVE N CLOSED (16)	Cannot detect or verify gate valve is closed.
ROUGH PUMP ERROR (32)	Roughing pump warning detected. See device properties.
TC CHILLER BURNOUT (64)	Chiller thermocouple unplugged or failed.
LN2 HEAT WARN (128)	Exhaust temperature on LN2 system nearing shutdown level.
GAUGE C WARNING (256)	Differential between absolute and Convector pressure gauges too great.
TC PUMP FLOW FAULT (512)	Temp differential between cold plate and coolant outlet out of range.
ETH FULL ALARM (1024)	Fluid level fill switch has been triggered.
VENT W ION GAUGE (2048)	Ion gauge won't shutdown and preventing venting.
OXY SENSOR OUT (4096)	LN2 oxygen sensor missing or nearing oxygen shutdown levels.
E STOP PUSHED (8192)	Emergency Stop is engaged.
RESERVED (16384)	Reserved
RECIRCULATOR WARNING (32768)	View heating & cooling recirculator device properties for warning code.
CYCLE TIME TOO SHORT (65536)	Huber fluid circulator not on for minimum cycle time

Table 8: Warning Codes

CODES: The current warning state of the system.

6.3 Service Notifications

6.3.1 GetExploraVacServiceNotifications

Usage: GetExploraVacServiceNotifications Response Body: {CODES}

Summary: Gets the current service issues on the system. Use SERVICEDEFINITIONS to decode.

CODES: The current service state of the system.

6.3.2 MarkExploraVacServiceReset

Usage: MarkExploraVacServiceReset {CODE} Response Body:

Code	Description
TIPSEAL (1)	Check/Replace roughing pump tip seals (24 months.).
PUMPBEARINGS (2)	Replace roughing pump bearings (60 months).
TURBOBEARINGS (4)	Service turbo pump bearings (48 months).
PLATEN ETHANOL HIGH (8)	Chiller reservoir overfilled.
PLATEN TCUBE COOLANT (16)	TCube coolant is low.
ROUGH VALVE (32)	Rebuild/replace pneumatic roughing line valve (1.5M cycles).
PURGE VALVE (64)	Rebuild/replace pneumatic purge line valve (1.5M cycles).
VENT VALVE (128)	Rebuild/replace pneumatic vent line valve (1.5M cycles).
FORELINE VALVE (256)	Rebuild/replace pneumatic foreline valve (1.5M cycles).
DELTAP (512)	Rebuild/replace Delta-P valve (500k cycles).
GATE VALVE (1024)	Rebuild/replace pneumatic gate valve (100k cycles).
ION GAUGE (2048)	Ion gauge needs degassing (3 months).
REPLACE ION (4096)	Ion gauge needs replaced (20k hours).
PLC BATTERY (8192)	PLC battery low on charge.
PLATEN ETHANOL LOW (16384)	Chiller fluid reservoir is low.
ROUGH COMMANDVALVE (32768)	Rebuild/replace roughing CommandValve (30M degrees).
PURGE COMMANDVALVE (65536)	Rebuild/replace purge CommandValve (30M degrees).
VENT COMMANDVALVE (131072)	Rebuild/replace vent CommandValve (30M degrees).
RECIRCULATOR SERVICE (262144)	View heating & cooling recirculator device properties for service code.

Table 9: Service Codes

Summary: Resets a single service issue on the system.

CODE: The service notification you want to mark as “serviced.”

7. Additional Notes

- Thermocouple setpoint and readout process values are rounded to the nearest 1/10th of a degree.
- Setpoint TTV values utilizing thermocouples are rounded to 1/100th of a degree.
- For simulated altitude calculation, NASA’s atmospheric model is used:

<https://www.grc.nasa.gov/www/k-12/airplane/atmosmet.html>